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Dog

The **dog** (***Canis familiaris*** or ***Canis lupus familiaris***) is a domesticated descendant of the wolf. Also called the **domestic dog**, it was domesticated from an extinct population of Pleistocene wolves over 14,000 years ago. The dog was the first species to be domesticated by humans. Experts estimate that hunter-gatherers domesticated dogs more than 15,000 years ago, which was before the development of agriculture. Due to their long association with humans, dogs have expanded to a large number of domestic individuals and gained the ability to thrive on a starch-rich diet that would be inadequate for other canids.[4]

The dog has been selectively bred over millennia for various behaviors, sensory capabilities, and physical attributes.[5] Dog breeds vary widely in shape, size, and color. They perform many roles for humans, such as hunting, herding, pulling loads, protection, assisting police and the military, companionship, therapy, and aiding disabled people. Over the millennia, dogs became uniquely adapted to human behavior, and the human–canine bond has been a topic of frequent study. This influence on human society has given them the sobriquet of "man's best friend".

Dog

Temporal range: Late Pleistocene to present[1]

Conservation status	
Domesticated	
Scientific classification	
Domain:	Eukaryota
Kingdom:	Animalia
Phylum:	Chordata
Class:	Mammalia
Order:	Carnivora
Family:	Canidae
Genus:	*Canis*
Species:	***C. familiaris***
Binomial name	
Canis familiaris Linnaeus, 1758[2]	
Synonyms[3]	
List [show]	

- *C. aegyptius* Linnaeus, 1758
- *C. alco* C. E. H. Smith, 1839,
- *C. americanus* Gmelin, 1792
- *C. anglicus* Gmelin, 1792

Taxonomy

In 1758, the Swedish botanist and zoologist Carl Linnaeus published in his *Systema Naturae*, the two-word naming of species (binomial nomenclature). *Canis* is the Latin word meaning "dog",[7] and under this genus, he listed the domestic dog, the wolf, and the golden jackal. He classified the domestic dog as *Canis familiaris* and, on the next page, classified the grey wolf as *Canis lupus*.[2] Linnaeus considered the dog to be a separate species from the wolf because of its upturning tail (*cauda recurvata* in Latin term), which is not found in any other canid.[8]

In 1999, a study of mitochondrial DNA (mtDNA) indicated that the domestic dog may have originated from the grey wolf, with the dingo and New Guinea

singing dog breeds having developed at a time when human communities were more isolated from each other.[9] In the third edition of *Mammal Species of the World* published in 2005, the mammalogist W. Christopher Wozencraft listed under the wolf *Canis lupus* its wild subspecies and proposed two additional subspecies, which formed the domestic dog clade: *familiaris*, as named by Linnaeus in 1758 and, *dingo* named by Meyer in 1793. Wozencraft included *hallstromi* (the New Guinea singing dog) as another name (junior synonym) for the dingo. Wozencraft referred to the mtDNA study as one of the guides informing his decision.[3] Mammalogists have noted the inclusion of *familiaris* and *dingo* together under the "domestic dog" clade[10] with some debating it.[11]

In 2019, a workshop hosted by the IUCN/Species Survival Commission's Canid Specialist Group considered the dingo and the New Guinea singing dog to be feral *Canis familiaris* and therefore did not assess them for the IUCN Red List of Threatened Species.[12]

Evolution

Domestication

The earliest remains generally accepted to be those of a domesticated dog were discovered in Bonn-Oberkassel, Germany. Contextual, isotopic, genetic, and morphological evidence shows that this dog was not a local wolf.[13] The dog was dated to 14,223 years ago and was found buried along with a man and a woman, all three having been sprayed with red hematite powder and buried under large, thick basalt blocks. The dog had died of canine distemper.[14] Earlier remains dating back to 30,000 years ago have been described as Paleolithic dogs, but their status as dogs or wolves remains debated[15] because considerable morphological diversity existed among wolves during the Late Pleistocene.[1]

This timing indicates that the dog was the first species to be domesticated[16][17] in the time of hunter-gatherers,[18] which predates agriculture.[1] DNA sequences show that all ancient and modern dogs share a common ancestry and descended from an ancient, extinct wolf population that was distinct from any modern wolf lineage. Some studies have posited that all living wolves are more closely related to each other than

- *C. antarcticus* Gmelin, 1792
- *C. aprinus* Gmelin, 1792
- *C. aquaticus* Linnaeus, 1758
- *C. aquatilis* Gmelin, 1792
- *C. avicularis* Gmelin, 1792
- *C. borealis* C. E. H. Smith, 1839
- *C. brevipilis* Gmelin, 1792
- *C. cursorius* Gmelin, 1792
- *C. domesticus* Linnaeus, 1758
- *C. extrarius* Gmelin, 1792
- *C. ferus* C. E. H. Smith, 1839
- *C. fricator* Gmelin, 1792
- *C. fricatrix* Linnaeus, 1758
- *C. fuillus* Gmelin, 1792
- *C. gallicus* Gmelin, 1792
- *C. glaucus* C. E. H. Smith, 1839
- *C. graius* Linnaeus, 1758
- *C. grajus* Gmelin, 1792
- *C. hagenbecki* Krumbiegel, 1950
- *C. haitensis* C. E. H. Smith, 1839
- *C. hibernicus* Gmelin, 1792
- *C. hirsutus* Gmelin, 1792
- *C. hybridus* Gmelin, 1792
- *C. islandicus* Gmelin, 1792
- *C. italicus* Gmelin, 1792
- *C. laniarius* Gmelin, 1792
- *C. leoninus* Gmelin, 1792
- *C. leporarius* C. E. H. Smith, 1839
- *C. lupus familiaris* Linnaeus,1758
- *C. major* Gmelin, 1792
- *C. mastinus* Linnaeus, 1758
- *C. melitacus* Gmelin, 1792
- *C. melitaeus* Linnaeus, 1758
- *C. minor* Gmelin, 1792
- *C. molossus* Gmelin, 1792
- *C. mustelinus* Linnaeus, 1758

to dogs,[19][18] while others have suggested that dogs are more closely related to modern Eurasian wolves than to American wolves.[20]

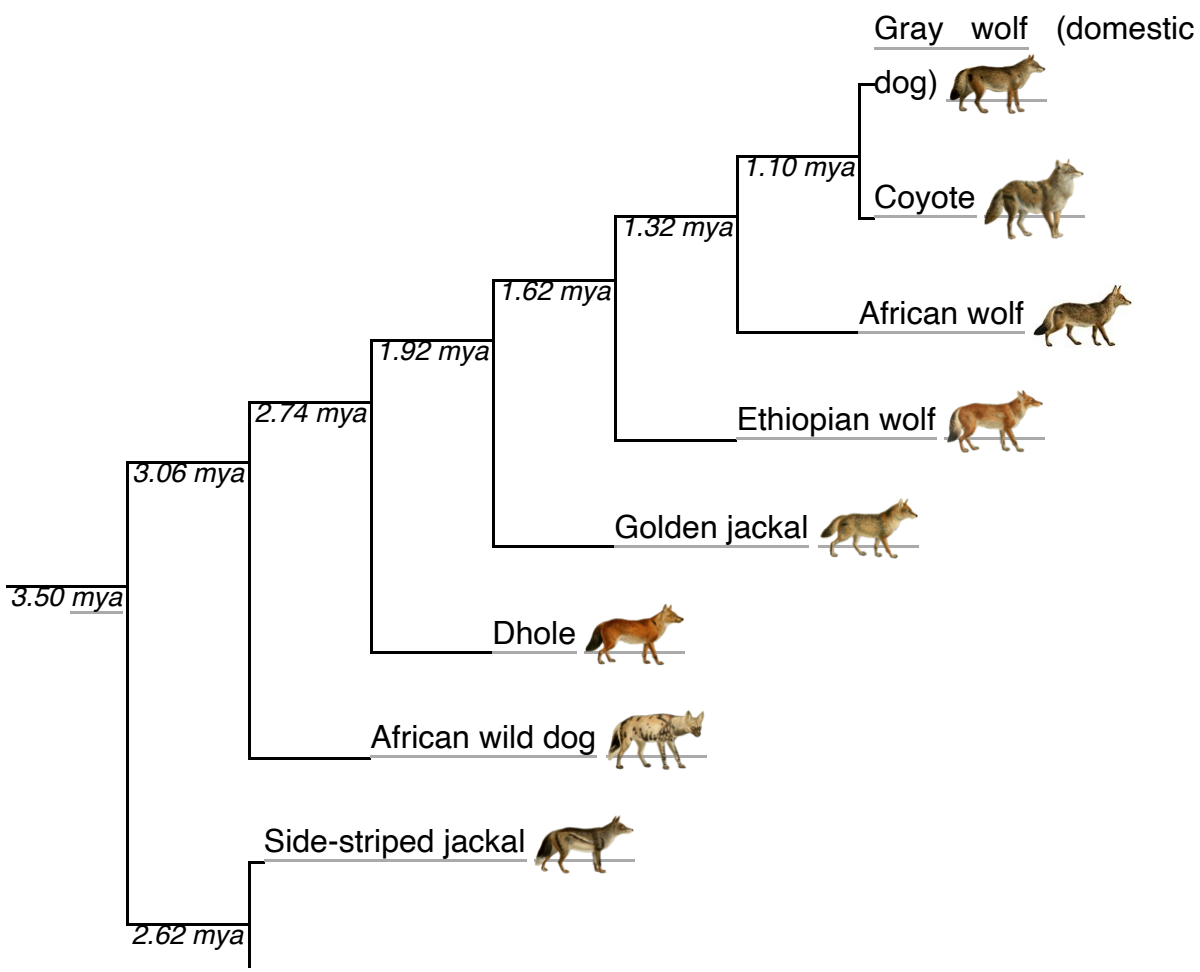
The dog is a classic example of a domestic animal that likely travelled a commensal pathway into domestication.[15][21] The questions of when and where dogs were first domesticated have taxed geneticists and archaeologists for decades.[16] Genetic studies suggest a domestication process commencing over 25,000 years ago, in one or several wolf populations in either Europe, the high Arctic, or eastern Asia.[22] In 2021, a literature review of the current evidence infers that the dog was domesticated in Siberia 23,000 years ago by ancient North Siberians, then later dispersed eastward into the Americas and westward across Eurasia,[13] with dogs likely accompanying the first humans to inhabit the Americas.[13] Some studies have suggested that the extinct Japanese wolf is closely related to the ancestor of domestic dogs.[23]

- *C. obesus* Gmelin, 1792
- *C. orientalis* Gmelin, 1792
- *C. pacificus* C. E. H. Smith, 1839
- *C. plancus* Gmelin, 1792
- *C. pomeranus* Gmelin, 1792
- *C. sagaces* C. E. H. Smith, 1839
- *C. sanguinarius* C. E. H. Smith, 1839
- *C. sagax* Linnaeus, 1758
- *C. scoticus* Gmelin, 1792
- *C. sibiricus* Gmelin, 1792
- *C. suillus* C. E. H. Smith, 1839
- *C. terraenovae* C. E. H. Smith, 1839
- *C. terrarius* C. E. H. Smith, 1839
- *C. turcicus* Gmelin, 1792
- *C. urcani* C. E. H. Smith, 1839
- *C. variegatus* Gmelin, 1792
- *C. venaticus* Gmelin, 1792
- *C. vertegus* Gmelin, 1792

Breeds

Dogs are the most variable mammal on earth, with around 450 globally recognized dog breeds.[22][24] In the Victorian era, directed human selection developed the modern dog breeds, which resulted in a vast range of

Canine phylogeny with ages of divergence



Black-backed jackal

Cladogram and divergence of the gray wolf (including the domestic dog) among its closest extant relatives[6]

phenotypes.[17] Most breeds were derived from small numbers of founders within the last 200 years.[17][22] Since then, dogs have undergone rapid phenotypic change and have been subjected to artificial selection by humans. The skull, body, and limb proportions between breeds display more phenotypic diversity than can be found within the entire order of carnivores. These breeds possess distinct traits related to morphology, which include body size, skull shape, tail phenotype, fur type and colour.[17] Their behavioural traits include guarding, herding, hunting,[17] retrieving, and scent detection. Their personality traits include hypersocial behavior, boldness, and aggression.[17] Present-day dogs are dispersed around the world.[22] An example of this dispersal is the numerous modern breeds of European lineage during the Victorian era.[18]

Location of a dog's carnassials; the inside of the 4th upper premolar aligns with the outside of the 1st lower molar, working like scissor blades.

Morphological variation in six dogs

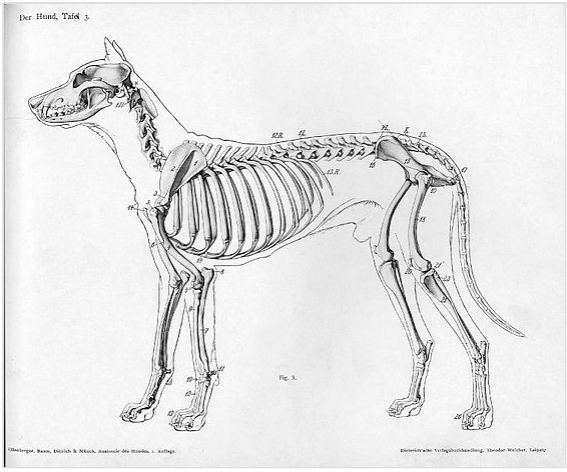
Phenotypic variation in four dogs

Anatomy and physiology

Size and skeleton

Dogs are extremely variable in size, ranging from one the largest breeds, the great dane, at 50 to 79 kg (110 to 174 lb) and 71 to 81 cm (28 to 32 in), to one of the smallest, the chihuahua, at 0.5 to 3 kg (1.1 to 6.6 lb) and 13 to 20 cm (5.1 to 7.9 in).[25][26] All healthy dogs, regardless of their size

and type, have an identical skeletal structure with the exception of the number of bones in the tail, although there is significant skeletal variation between dogs of different types.[27][28] The dog's skeleton is well adapted for running; the vertebrae on the neck and back have extensions for back muscles, consisting of epaxial muscles and hypaxial muscles, to connect to; the long ribs provide room for the heart and lungs; and the shoulders are unattached to the skeleton, allowing for flexibility.[27][28][29]



A lateral view of a dog skeleton

Compared to the dog's wolf-like ancestors, selective breeding since domestication has seen the dog's skeleton larger in size for larger types such as mastiffs and miniaturised for smaller types such as terriers; dwarfism has been selectively used for some types where short legs are advantageous, such as dachshunds and corgis.[28] Most dogs naturally have 26 vertebrae in their tails, but some with naturally short tails have as few as three.[27]

The dog's skull has identical components regardless of breed type, but there is significant divergence in terms of skull shape between types.[28][30] The three basic skull shapes are the elongated dolichocephalic type as seen in sighthounds, the intermediate mesocephalic or mesaticephalic type, and the very short and broad brachycephalic type exemplified by mastiff type skulls.[28][30] The jaw contains around 42 teeth, and is designed for the consumption of flesh. Dogs use their carnassial teeth to cut food into bite-sized chunks, more especially meat.[31]

Senses

Dogs' senses include vision, hearing, smell, taste, touch, and magnetoreception. One study suggests that dogs can feel small variations in Earth's magnetic field.[32] Dogs prefer to defecate with their spines aligned in a north-south position in calm magnetic field conditions.[33]

Dogs' vision is dichromatic; the dog's visual world consists of yellows, blues, and grays. They have difficulty differentiating between red and green. The divergence of the eye axis of dogs ranges from 12–25°, depending on the breed. Dogs' eyes of different breeds can have different retina configurations. The fovea centralis area of dogs' eyes, which is attached to a nerve fiber, is the most sensitive to photons.[34]

Coat

The coats of domestic dogs are of two varieties: "double" being familiar with dogs (as well as wolves) originating from colder climates, made up of a coarse guard hair and a soft down hair, or "single", with the topcoat only. Breeds may have an occasional "blaze", stripe, or "star" of white fur on their chest or underside.[35] Premature graying can occur in dogs as early as one year of age; this is associated with impulsive behaviors, anxiety behaviors, and fear of unfamiliar noise, people, or animals.[36] Some dog breeds are hairless, while others have a very thick corded coat. The coats of certain breeds are often groomed to a characteristic style, for example, the Yorkshire terrier's "show cut".[31]

Dewclaw